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EXPERIMENT 3

Aim:

1. To load a large dataset into a data warehouse and run queries on it. (e.g. Google or BigQuery).
2. To implement SQL-based transformations using data build tool to build a reliable data transformation online.

Theory:

How to Load a large dataset into a data warehouse and run queries on it. (e.g., Google or BigQuery).

Step 1: Preparing the Dataset

Before loading the dataset into a data warehouse, ensure:

- The dataset is in a supported format (CSV, JSON, Avro, ORC, Parquet).
- The data is **clean** (no missing or corrupted values).
- The schema (columns, data types) is well-defined.

For BigQuery:

- The **file size limit** for a single load job is **5 TB**.
- If using Google Cloud Storage (GCS), ensure files are compressed (GZIP, Snappy) to save space.

Step 2: Uploading the Dataset

Option 1: Upload Directly from Local Storage

- You can upload files directly through the **BigQuery Web UI** or using the **bq command-line tool**.

Option 2: Upload from Google Cloud Storage

- If the dataset is large, upload it first to **Google Cloud Storage (GCS)**.

Step 3: Optimizing Query Performance

Once the data is loaded, follow these best practices for running efficient queries:

1. Partitioning the Table



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- Partition the table based on **date** or **range** to improve query performance.

2. Clustering the Table

- Cluster on frequently filtered columns.

3. Running Queries Efficiently

- Use **SELECT specific columns** instead of SELECT * to reduce scan costs.
- Use **approximate functions** for large datasets (APPROX_COUNT_DISTINCT() instead of COUNT(DISTINCT column)).
- Use **filters early** in the query (WHERE clause) to reduce data scanning.

Step 4: Automating Data Loading (ETL Pipeline)

For automated and scheduled data loading:

- Use **Cloud Dataflow (Apache Beam)** for complex ETL tasks.
- Use **Cloud Composer (Airflow)** to schedule ETL pipelines.
- Use **BigQuery scheduled queries** for periodic transformations.

Screenshots:

Upload Data from GUI to perform Big Query:

For Csv file format:



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Add

Source

Search for data sources

Popular sources

Local file

Upload a local file

Google Cloud Storage

Google object storage service

Connections to external data sources

Connection from BigQuery to an external data source

PREVIEW

Auto-create external and BigLake tables from Cloud Storage

Automatically scan Cloud Storage buckets to create and update BigLak...

Additional sources

Viewing all 31 results.

Search for and star a project

Search for a BigQuery project and add it to the Explorer

Star a project by name

Add a BigQuery project to the Explorer by project name

Analytics Hub

Discover and subscribe to public, commercial or privately shared data sets

Google Drive

Google storage service

Salesforce Data Cloud

CLOSE

Create table

Source

Create table from

Upload

Select file *

emp.csv

X BROWSE ?

File format

CSV

Destination

Project *

dataeng09

BROWSE

Data set *

ds1

Table *

emp

Maximum name size is 1,024 UTF-8 bytes. Unicode letters, marks, numbers, connectors, dashes and spaces are allowed.

Table type

Native table

?

Schema

Auto-detect

Schema will be automatically generated.

Partitioning settings

No partitioning

Partition by ingestion time

Partition by field

To partition by field, you must first add fields to the schema above.

CREATE TABLE

CANCEL

Data shown from that CSV:



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The screenshot shows the Google Cloud BigQuery console interface. On the left is a navigation menu with options like Pipelines and Integration, Data transfers, Pipelines (Dataform), Scheduled queries, and Governance. The main area displays a table named 'data' with the following columns: Row, Emp ID, Name Prefix, First Name, Middle Initial, Last Name, Gender, and E Mail. The table contains 50 rows of employee data. Below the table, there is a 'Job history' section.

Perform Query on that csv data:

The screenshot shows the Google Cloud BigQuery console interface. The main area displays a query result for the query 'SELECT * FROM `dataeng09`.`del\_data` LIMIT 1000'. The results show a list of employees with the following columns: Row, Emp ID, Name Prefix, First Name, Middle Initial, Last Name, Gender, and E Mail. The table contains 25 rows of employee data.

For ORC file format:



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Create table

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Source

Create table from Upload

Select file *
orc1.orc X BROWSE ?

File format
ORC ?

Destination

Project *
dataeng09 BROWSE

Data set *
de1

Table *
orc

Maximum name size is 1,024 UTF-8 bytes. Unicode letters, marks, numbers, connectors, dashes and spaces are allowed.

Table type
Native table ?

Schema

! Source file defines the schema.

Partitioning settings

- ☒ No partitioning
- ☐ Partition by ingestion time
- ☐ Partition by field
To partition by field, you must first add fields to the schema above.

CREATE TABLE CANCEL

The screenshot shows the Google Cloud BigQuery console interface. The left sidebar contains navigation options like Studio, Pipelines and Integration, Governance, Administration, and Migration. The main area displays the 'orc' table schema with columns: boolean1, byte1, short1, int1, long1, float1, double1, bytes1, string1, middle, list, map, ts, and decimal1. The table is currently empty. The bottom status bar shows the time as 10:53 on 18-02-2025.



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The screenshot shows the Google Cloud BigQuery console. The Explorer view displays a dataset named 'orc' under the project 'dataeng09'. The dataset contains 12 tables, each with 1024 rows. The tables are named 'orc_1' through 'orc_12'. The 'orc' dataset is selected, and its details are shown in the 'DETAILS' tab. The 'TABLE EXPLORER' tab shows the schema of the 'orc' dataset, which includes columns: 'boolean', 'byte1', 'short1', 'int1', 'long1', 'float1', 'double1', 'bytes1', 'string1', and 'mode1'. The 'DATA QUALITY' tab shows the data quality metrics for the 'orc' dataset, including 'mode1' and 'string1'.

Row	boolean	byte1	short1	int1	long1	float1	double1	bytes1	string1	mode1
1	false	1	1024	65536	922337203...	1.0	-15.0	AAECAwQ+	Ni	1
2	false	1	1024	65536	922337203...	1.0	-15.0	AAECAwQ+	Ni	2
3	false	1	1024	65536	922337203...	1.0	-15.0	AAECAwQ+	Ni	1
4	false	1	1024	65536	922337203...	1.0	-15.0	AAECAwQ+	Ni	2
5	false	1	1024	65536	922337203...	1.0	-15.0	AAECAwQ+	Ni	1
6	false	1	1024	65536	922337203...	1.0	-15.0	AAECAwQ+	Ni	2
7	false	1	1024	65536	922337203...	1.0	-15.0	AAECAwQ+	Ni	1
8	false	1	1024	65536	922337203...	1.0	-15.0	AAECAwQ+	Ni	2
9	false	1	1024	65536	922337203...	1.0	-15.0	AAECAwQ+	Ni	1
10	false	1	1024	65536	922337203...	1.0	-15.0	AAECAwQ+	Ni	2
11	false	1	1024	65536	922337203...	1.0	-15.0	AAECAwQ+	Ni	1
12	false	1	1024	65536	922337203...	1.0	-15.0	AAECAwQ+	Ni	2

Perform Query on ORC file:

The screenshot shows the Google Cloud BigQuery console with an 'Untitled query' view. The query executed is 'SELECT * FROM `dataeng09`.`orc` LIMIT 1000'. The query results are displayed in a table with columns: 'boolean', 'byte1', 'short1', 'int1', 'long1', 'float1', 'double1', 'bytes1', 'string1', and 'mode1'. The results show 8 rows of data, each with a 'boolean' value of 'false', a 'byte1' value of '1', and a 'short1' value of '1024'. The 'int1' column contains values like '65536' and '9223372036854...'. The 'float1' column contains values like '1.0' and '-15.0'. The 'bytes1' column contains the value 'AAECAwQ+'. The 'string1' column contains the value 'Ni'. The 'mode1' column contains the value '1'.

Row	boolean	byte1	short1	int1	long1	float1	double1	bytes1	string1	mode1
1	false	1	1024	65536	9223372036854...	1.0	-15.0	AAECAwQ+	Ni	1
2	false	1	1024	65536	9223372036854...	1.0	-15.0	AAECAwQ+	Ni	2
3	false	1	1024	65536	9223372036854...	1.0	-15.0	AAECAwQ+	Ni	1
4	false	1	1024	65536	9223372036854...	1.0	-15.0	AAECAwQ+	Ni	2
5	false	1	1024	65536	9223372036854...	1.0	-15.0	AAECAwQ+	Ni	1
6	false	1	1024	65536	9223372036854...	1.0	-15.0	AAECAwQ+	Ni	2
7	false	1	1024	65536	9223372036854...	1.0	-15.0	AAECAwQ+	Ni	1
8	false	1	1024	65536	9223372036854...	1.0	-15.0	AAECAwQ+	Ni	2

Bucket Creation and Upload data in it:



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Google Cloud

My First Project

bucket

Cloud Storage

Create a Bucket

Overview

Buckets

Monitoring

Settings

Get started

Name: bucket_tushar_jcb

Choose where to store your data

Location: asia-south1 (Mumbai)

Location type: Region

Choose a storage class for your data

Default storage class: Standard

Choose how to control access to objects

Public access prevention: Off

Access control: Uniform

Choose how to protect object data

Your data is always protected with Cloud Storage but you can also choose from these additional data protection options to add extra layers of security.

Data protection

Soft-delete policy (for data recovery)

When enabled, this bucket and its objects will be kept for a specified period after they're deleted and can be restored during this time. [Learn more](#)

Use default retention duration

All buckets have a seven-day soft delete duration by default, unless this default has been customised by your organisation administrator.

Set custom retention duration

Specify how long this bucket and its objects should be retained after they're deleted. Setting a '0' duration disables soft delete, meaning that any deleted objects will be deleted permanently.

Object versioning (for version control)

For restoring deleted or overwritten objects. To minimise the cost of storing versions, we recommend limiting the number of non-current versions per object and scheduling them to expire after a number of days. [Learn more](#)

Retention (for compliance)

For preventing the deletion or modification of the bucket's objects for a specified period of time.

DATA ENCRYPTION

CREATE CANCEL

Good to know

Location pricing

Storage rates vary depending on the storage class of your data and location of your bucket. [Pricing details](#)

Current configuration: Region / Standard

Item	Cost
asia-south1 (Mumbai)	\$0.023 per GB/month

ESTIMATE YOUR MONTHLY COST

bucket_tushar_jcb - Bucket details

Inbox (889) - tpancha2035@gmail.com

console.cloud.google.com/storage/browser/bucket_tushar_jcb?tab=objects?project=festive-quanta-451305-g4&prefix=&forceOnObjectsSortingFiltering=false

Google Cloud

My First Project

bucket

Search

Cloud Storage

Bucket details

GO TO PATH REFRESH LEARN

Overview

Buckets

Monitoring

Settings

bucket_tushar_jcb

Location: asia-south1 (Mumbai)

Storage class: Standard

Public access: Not public

Protection: Soft delete

OBJECTS CONFIGURATION PERMISSIONS PROTECTION LIFECYCLE OBSERVABILITY INVENTORY REPORTS OPERATIONS

Folder browser

bucket_tushar_jcb

CREATE FOLDER UPLOAD TRANSFER DATA OTHER SERVICES

Filter by name prefix only Filter Filter objects and folders

Name	Size	Type	Created	Storage class	Last modified	Public access	Version history	Encryption	Object retention retain until time	Retention expiry time
No rows to display										

your bucket is ready. Just add data.



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Add



Source

Search for data sources

Popular sources



Local file

Upload a local file



Google Cloud Storage

Google object storage service



Connections to external data
sources

Connection from BigQuery to an
external data source



PREVIEW

Auto-create external and BigLake
tables from Cloud Storage

Automatically scan Cloud Storage
buckets to create and update BigLake...

Create table

Source

Create table from
Google Cloud Storage

Select file from GCS bucket or [use a URI pattern](#)

File format
Parquet

☐ Source data partitioning

Destination

Project *
festive-quanta-451305-g4

Data set *

Table *

Maximum name size is 1,024 UTF-8 bytes. Unicode letters, marks, numbers, connectors, dashes and spaces are allowed.

Table type
Native table

Schema

Source file defines the schema.

Partitioning settings

☒ No partitioning
☐ Partition by ingestion time
☐ Partition by field
To partition by field, you must first add fields to the schema above.

CREATE TABLE CANCEL

Choose a file

bucket_tushar_icb



mtcars.parquet

Filename
mtcars.parquet

SELECT CANCEL



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Create table

Source

Create table from
Google Cloud Storage

Select file from GCS bucket or use a URI pattern [?](#) *
☒ bucket_tushar_icb/mtcars.parquet

BROWSE ?

File format
Parquet

☐ Source data partitioning

Destination

Project *
festive-quanta-451305-g4

BROWSE

Data set *
bucket_1

Table *
slide_1
Maximum name size is 1,024 UTF-8 bytes. Unicode letters, marks, numbers, connectors, dashes and spaces are allowed.

Table type
Native table

Schema

Source file defines the schema.

Partitioning settings

☒ No partitioning
☐ Partition by ingestion time
☐ Partition by field
To partition by field, you must first add fields to the schema above.

CREATE TABLE

CANCEL

bucket_tushar_icb

Location
asia-south1 (Mumbai)

Storage class
Standard

Public access
Not public

Protection
Soft delete

OBJECTS

CONFIGURATION

PERMISSIONS

PROTECTION

LIFECYCLE

OBSERVABILITY

INVENTORY REPORTS

OPERATIONS

Folder browser

bucket_tushar_icb

CREATE FOLDER

UPLOAD

TRANSFER DATA

OTHER SERVICES

Filter by name prefix only

Filter

Filter objects and folders

Show Live objects only

☐	Name	Size	Type	Created	Storage class	Last modified	Public access	Version history	
☐	mtcars.parquet	4.5 KB	application/octet-stream	18 Feb 2025, 11:34:56	Standard	18 Feb 2025, 11:34:56	Not public	—	⬇ ⋮



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Row	model	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
1	Merc 240D	24.4	4	146.7	62	3.69	3.19	20.0	1	0	4	2
2	Merc 230	22.8	4	140.8	95	3.92	3.15	22.9	1	0	4	2
3	Toyota Corona	21.5	4	120.1	97	3.7	2.465	20.01	1	0	3	1
4	Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
5	Valiant	18.1	6	225.0	105	2.76	3.46	20.22	1	0	3	1
6	Merc 280	19.2	6	167.6	123	3.92	3.44	18.3	1	0	4	4
7	Merc 280C	17.8	6	167.6	123	3.92	3.44	18.9	1	0	4	4
8	Hornet Sportabout	18.7	8	360.0	175	3.15	3.44	17.02	0	0	3	2
9	Duster 360	14.3	8	360.0	245	3.21	3.57	15.84	0	0	3	4
10	Merc 450SE	16.4	8	275.8	180	3.07	4.07	17.4	0	0	3	3
11	Merc 450SE	17.3	8	275.8	180	3.07	3.73	17.6	0	0	3	3
12	Merc 450SE C	15.2	8	275.8	180	3.07	3.78	18.0	0	0	3	3
13	Cadillac Fleetwood	10.4	8	472.0	205	2.93	5.25	17.98	0	0	3	4
14	Lincoln Continental	10.4	8	460.0	215	3.0	5.424	17.82	0	0	3	4
15	Chrysler Imperial	14.7	8	440.0	230	3.23	5.345	17.42	0	0	3	4
16	Dodge Challenger	15.5	8	318.0	150	2.76	3.52	16.87	0	0	3	2
17	AMC Javelin	15.2	8	304.0	150	3.15	3.435	17.3	0	0	3	2
18	Camaro Z28	13.3	8	390.0	245	3.73	3.84	15.41	0	0	3	4
19	Pontiac Firebird	19.2	8	400.0	175	3.08	3.845	17.05	0	0	3	2
20	Datsun 710	22.8	4	108.0	93	3.85	2.32	18.61	1	1	4	1
21	Fiat 128	32.4	4	78.7	66	4.08	2.2	19.47	1	1	4	1
22	Honda Civic	30.4	4	75.7	52	4.93	1.615	18.52	1	1	4	2
23	Toyota Corolla	33.9	4	71.1	65	4.22	1.835	19.9	1	1	4	1
24	Fiat X1-9	27.3	4	79.0	66	4.08	1.935	18.9	1	1	4	1
25	Porsche 914-2	26.0	4	120.3	91	4.43	2.14	16.7	0	1	5	2
26	Lotus Europa	30.4	4	95.1	113	3.77	1.513	16.9	1	1	5	2
27	Volvo 740 GLE	11.4	8	517.0	300	4.11	5.78	19.4	1	1	4	2

Perform Query on that Bucket Data:

Query: `SELECT * FROM 'festive-quanta-451305-g4.bucket_1.slide_1' LIMIT 1000`

Row	model	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
1	Merc 240D	24.4	4	146.7	62	3.69	3.19	20.0	1	0	4	2
2	Merc 230	22.8	4	140.8	95	3.92	3.15	22.9	1	0	4	2
3	Toyota Corona	21.5	4	120.1	97	3.7	2.465	20.01	1	0	3	1
4	Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
5	Valiant	18.1	6	225.0	105	2.76	3.46	20.22	1	0	3	1
6	Merc 280	19.2	6	167.6	123	3.92	3.44	18.3	1	0	4	4
7	Merc 280C	17.8	6	167.6	123	3.92	3.44	18.9	1	0	4	4
8	Hornet Sportabout	18.7	8	360.0	175	3.15	3.44	17.02	0	0	3	2
9	Duster 360	14.3	8	360.0	245	3.21	3.57	15.84	0	0	3	4
10	Merc 450SE	16.4	8	275.8	180	3.07	4.07	17.4	0	0	3	3
11	Merc 450SE	17.3	8	275.8	180	3.07	3.73	17.6	0	0	3	3
12	Merc 450SE C	15.2	8	275.8	180	3.07	3.78	18.0	0	0	3	3
13	Cadillac Fleetwood	10.4	8	472.0	205	2.93	5.25	17.98	0	0	3	4

Create Pipeline on Google Cloud:



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[←](#) Create repository

Repositories contain a single Dataform project that can be connected to your Git provider. Within a repository, you can create workspaces for development and execute your SQL workflows against BigQuery.

Repository ID *

tushar_001

Region *

asia-south1 (Mumbai)

Service account

Default Dataform service account

☐ Use the default KMS key for encryption

Encryption
Google-managed encryption key will be ignored if the default customer-managed encryption key is set in this project location.
☒ Google-managed encryption key
Keys owned by Google
☐ Cloud KMS key
Keys owned by customers

New repositories start empty and can be connected to a git provider after being created.

Creating...

Google Cloud My First Project bucke

🔍

← Create repository

📁

•

⚙️

➡️

🕒

⚙️

🔍

🔒

•

•

⚙️

🔧

✅ Repository created successfully.

🔔 Dataform will execute workflows as the service account service-866073429415@gcp-sa-dataform.iam.gserviceaccount.com.

⚠️ To execute queries in Dataform, you need the following roles:
for service-866073429415@gcp-sa-dataform.iam.gserviceaccount.com:

- roles/bigquery.jobUser on festive-quanta-451305-g4

Grant all Learn more

Go to repositories



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The screenshot shows the Google Cloud Dataform console. The left sidebar contains navigation options: Studio, Pipelines & Integration, Data transfers, Pipelines (Dataform), Scheduled queries, Scheduling, Governance, Analytics Hub, Policy tags, Administration, Migration, Assessment, SQL translation, Partner Center, Settings, and Release Notes. The main panel displays the 'Filter repositories' table with columns: Name, Location, Source, Authentication status, and Action. A single repository 'tushar_001' is listed with location 'asia-south1'. The right sidebar shows 'Recommended for you' with links to get started, create and execute a workflow, schedule production executions, create development environments, and learn more.

The screenshot shows the Google Cloud Dataform console for the 'tushar_001' repository. The main panel displays the 'Development Workspaces' section with a table listing workspaces. One workspace, 'tushar_devarsh_01', is listed. The right sidebar shows 'Recommended for you' with links to create a workspace, create a release configuration, create a workflow configuration, monitor execution logs, and configure workspace compilation overrides.

The screenshot shows the Google Cloud Dataform console displaying a compiled graph. The graph consists of two nodes: 'first_view' and 'second_view', both with the view name 'tushar_devarsh_01'. The 'first_view' node is connected to the 'second_view' node. The right sidebar shows the 'Query' tab with a sample SQL query:

```
1 -- This is an example SQLX file to help you learn the basics of Dataform.
2 -- Visit https://cloud.google.com/dataform/docs/sql-workflows for more information on how to configure your SQL workflow.
3
4 -- You can delete this file, then commit and push your changes to your repository when you are ready.
5
6 -- Config blocks allow you to configure, document, and test your data assets.
7
8
9 -- The rest of a SQLX file contains your SELECT statement used to create the table.
10
11 SELECT 1 as test
```

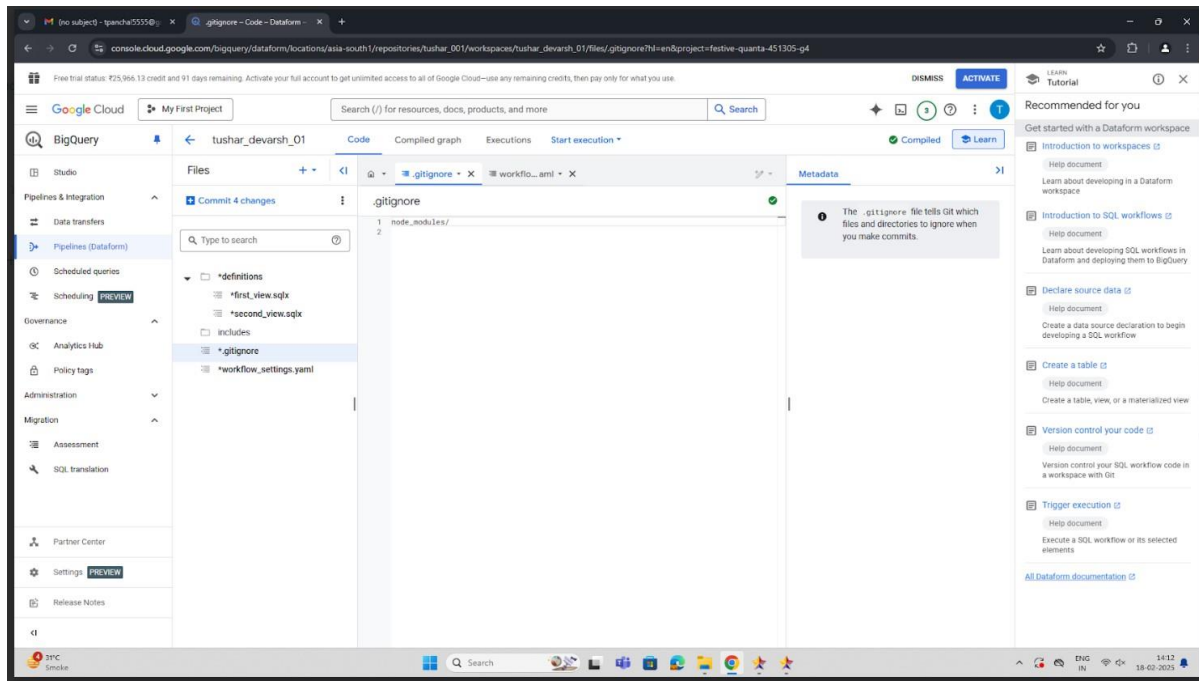


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Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)



Schedule Query:



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Export as scheduled query

✔ Settings valid.

Details and schedule

Name for scheduled query *

count_all_q

Schedule options

Repeat frequency *

Minutes

Repeat frequency *

5

minutes



☒ Start now ☐ Start at set time

Start date and run time

18/02/2025, 11:45

IST



☐ End never ☒ Schedule end time

End date *

19/02/2025, 00:00

IST



Scheduled to run every 5 minutes starting now and ending on 19 February 2025 at 00:00:00 IST.

Destination for query results

☒ Set a destination table for query results

Dataset *

☒ festive-quanta-451305-g4.bucket_1

Table Id *

scheduled_time

Destination table partitioning field

Destination table write preference

☒ Append to table

☐ Overwrite table

SAVE

CANCEL



**SHRI VILEPARLE KELAVANI MANDAL'S
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING**
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The screenshot shows the Google Cloud BigQuery console interface. The top navigation bar includes the Google Cloud logo, the project name 'My First Project', and a search bar. The left sidebar contains a navigation menu with options like 'Show starred only', 'Queries', 'Notebooks', 'Data canvases', 'Data preparations', 'Workflows', 'External connections', and 'Connections'. The main area displays the query 'count_all_q' with its SQL code:

```
1 SELECT COUNT(*)
2 FROM `festive-quanta-451305-g4.bucket_1.slide_1`;
3
```

 Below the query, the 'Query results' section shows a table with one row containing the value '32'. The 'Job history' section lists several jobs, including the current one 'count_all_q' which is marked as 'Succeeded'.

Scheduled queries							
+ CREATE SCHEDULED QUERY IN EDITOR							
DELETE							
Filter Filter scheduled query configs							
☐	☒	Display name	Source	Schedule (UTC)	Region	Destination data set	Next scheduled UTC+5:30
☐	☒	count_all_q	Scheduled Query	every 5 minutes	asia-south1	bucket_1	18 February 2025 at 12:00:00 UTC+5:30

count_all_q				
Succeeded				
e (UTC) Target date for next run				
every 5 minutes 18 February 2025 at 12:00:00 UTC+5:30				
Filter Filter scheduled query runs				
☒	Run date	Schedule time	UTC+5:30	Summary
☒	18 February 2025	18 February 2025 at 11:55:00 UTC+5:30		The scheduled query has completed successfully.

Conclusion

Thus we successfully implemented loading a large dataset into a data warehouse and ran queries on it and implemented SQL-based transformations using data build tool to build a reliable data transformation online.